

IAL Biology Unit 6 (WBI16): Rapid Revision

Practical Skills in Biology II · last-minute summary · 1 h 20 min · 50 marks · 4 questions · planning question 15–18 marks · 1.6 min per mark · no statistical test to derive from scratch

HOW THE MARKS WORK

- Nearly every extended answer is marked as a **list of independent points** — “an answer that includes **eight** of the following”. Write short, separate, numbered sentences, one idea each.
- You have **1.6 min per mark**, the most generous paper in the qualification. Use it to write every method point.
- Quantities, not “amount”**: volume, mass, number, with a unit.
- Stated numbers**: “0, 0.5, 1.0, 1.5 and 2.0 mol dm⁻³”, never “different concentrations”.
- Say **how** a variable is controlled, not just which.
- Effects must be **directional**.

THE PAPER

Q	Marks	Content
1–2	~10 ea	Method: describe an experiment, measure something accurately, criticise recorded data
3	~14	Data analysis: table, graph, a statistical test, a conclusion
4	15–18	The planning question

THE PLANNING QUESTION

Part	Marks	What scores
(a)	2–3	Preliminary work. Always a “ find a suitable... ” statement — never a method. Say what the “suitable” is for : “a suitable temperature for measurable respiration to occur ”.
(b)	8–10	The method. Every separate correct point is a mark.
(c)	3–4	Table + graph + named test.
(d)	2–4	Limitations — one of each of the three kinds.

THE METHOD: THE UNIVERSAL MARK POINTS

Write one sentence each.

- DV stated clearly with how it is measured** — “volume of O₂ released per minute”. Ignore “amount”; ignore “rate” alone. **Do not piece together the dependent variable.**
- IV with a **range of stated values**.
- How the IV is changed.
- Apparatus for measuring the DV (respirometer, photosynthometer, colorimeter; a labelled diagram is accepted).
- Standardise the organisms** — same mass / length / strain / source.
- Acclimatise or equilibrate** before measuring.
- A **stated time** for the measurement.
- Calculate a rate** — volume ÷ time, 1 ÷ time, $\pi r^2 d \div t$.

- Two controlled variables and how each is controlled.**
- A **control** for comparison.
- Aseptic technique if microbes.
- Repeats** → **mean AND standard deviation**.
- Welfare if animals.

Limitations — give one of each kind

- Measuring the DV is hard: small gas volumes; O₂ dissolves in water; judging a colour end point; zones not circular.
- A variable is hard to control: germinating seeds and lamps produce heat; age, viability or genetic variation.
- The biology changes during the experiment: aerobic → anaerobic; waste builds up; gases expand when warm; CO₂ is soluble.

Ignore bubbles or escape of gas unqualified — say **what** is hard and **why it matters**.

VARIABLES, CONTROL AND RISK

THE THREE-PART VARIABLE QUESTION

3–5 marks, every paper.

- Name the variable in the **right category**. A biotic answer on an abiotic line scores 0 and **negates the line**. Write **one variable per line** — only the first answer on each line is marked.
- Say **how** it is controlled — the method, not the variable.
- The effect if uncontrolled **must be directional**. “It would affect the results” = 0. “A higher temperature would increase enzyme activity, **so more CO₂ would be produced**” = 1.

The safe fallback, always on the list: “**the results would not be valid**”. Never write “not reliable” or “not accurate” — both are ignored.

Variable	Control method the schemes accept
Temperature (solution)	Thermostatic water bath or incubator — ignore “water bath” unqualified
Temperature (room, animal)	Air-conditioned or thermostatically controlled room
Temperature (aquatic plant)	Thermostatic water bath; heat shield between lamp and plant
pH	Buffer at a stated pH; in the field, select sites of similar pH
Light intensity / wavelength	Same wattage bulb at a fixed measured distance ; light meter; same filter
Mass / length of material	Balance; ruler; cork borer — ignore “size”
Age of organism	Hatched at the same time; seeds from the same batch
Stimulus (touch)	Same glass rod, same force, same person

Things the schemes ignore as variables: “nutrients” (say minerals, nitrate, glucose), “size”, “type”, “genetic variation”, “stress”.

RISK = A NAMED RISK + A MATCHING REDUCTION

Microbiology: infection by pathogens — not “contamination” unqualified — reduced by aseptic technique, taping the lid, incubating below 30 °C. Spirometer: infection from the mouthpiece — sterilise or replace it. Fibre tensile test: snapping fibre or falling masses — goggles and a sand tray. **“Wear a lab coat” and “be careful” score nothing.**

TABLE, GRAPH AND THE FOUR TESTS

THE PAIRING RULE

A mismatch costs the mark.

- **Two groups compared** (two varieties, with vs without, trained vs untrained) → **bar graph + t-test** (difference).
- **Several values of a continuous IV** (5 light intensities, 5 temperatures, ages) → **line or scatter graph + Spearman’s rank** (correlation).
- **Counts in categories** (choice chamber, fruiting / not fruiting) → **bar chart + chi-squared**.

The table — 3 marks

Suitable format with raw data · **headings with units, unit in the heading only** (do not accept units in the body) · **means to the same number of decimal places as the data. An extra column negates the first mark** — no “trial number” column, no calculated rate in a raw-data table.

Test	Formula and decision
Student’s t (12 of 19 papers)	$t = \bar{x}_1 - \bar{x}_2 \div \sqrt{s_1^2/n_1 + s_2^2/n_2}$ df = $n_1 + n_2 - 2$ $t > \text{critical} \rightarrow$ reject H_0 : significant difference. The paper gives you s^2 — do not square it again. Ignore the minus sign.
Spearman’s rank (5 papers)	$r_s = 1 - \frac{6\sum d^2}{n(n^2 - 1)}$ Rank each variable separately; ties share the mean rank. $r_s > \text{critical} \rightarrow$ reject H_0 : significant correlation. State positive or negative. At $n = 5$ the critical value is 1.000 — only a perfect correlation is significant.
Chi-squared (3 papers)	$\chi^2 = \Sigma(O - E)^2 \div E$ df = categories – 1; E = total ÷ number of categories. $\chi^2 > \text{critical} \rightarrow$ reject H_0 . Used because the data are categorical frequency counts .
Mann-Whitney U (2 papers)	$U = R - n(n+1)/2$ for each group; R = sum of ranks when both groups are ranked together. Opposite way round: significant if the smallest U is ≤ the critical value . Compares medians .

THE DECISION SENTENCE — NEVER OMIT THE COMPARISON

“The calculated value of ____ is **greater / less than** the critical value of ____ at $p = 0.05$, **so we reject / accept the null hypothesis**: there is (no) significant difference / correlation between ____ and ____.”

“Must state the calculated value is more or less than the critical value” — a decision without the comparison scores **nothing**.

Null hypothesis = there is **no significant difference or correlation** between the two named things.

Error bars and other calculations

Overlap → not significant; no overlap → significant. Always name which groups — do not accept “overlapping SDs are not significant” unqualified. Add “a statistical test would be needed to confirm”.

% change = $(\text{new} - \text{original}) \div \text{original} \times 100$ · magnification = $\text{image} \div \text{actual}$ (convert units first) · rate = $\text{change} \div \text{time}$ **with units** (write $\text{dm}^3 \text{s}^{-1}$, not dm^3/s).

THE REJECTIONS BANK

Do not write	Write instead
"Amount"; "different temperatures"	Volume / mass / number with a unit; stated values
"It would affect the results"	A directional effect on the DV
"Water bath" unqualified	Thermostatic water bath
"Be careful"; "wear a lab coat"	A specific reduction matched to the named risk
Units in the body of the table; an extra column	Units in the heading only ; no trial-number column
A decision with no comparison to the critical value	"X is greater than the critical value of Y, so reject H_0 "
"Not reliable" / "not accurate"	"The results would not be valid "

WALKING INTO THE EXAM

You have 1.6 min per mark, so write every method point. State numbers. Say **how** each variable is controlled and give a **directional** effect. Match the graph to the test. Never give a decision without comparing to the critical value.